BUILDING REGULATIONS POSITION STATEMENT

Dual Pumps, Dual Sump Chambers & Battery Backup
Property: 1 Riverside Cottages, The Dean, Alresford, Hampshire, SO24 9BE

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1. Purpose

This document sets out why the provision of **dual sump pumps, separate internal and external sump chambers, and battery backup** is **required to achieve compliance with the Building Regulations**, and does not constitute betterment.

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2. Building Regulations – Functional Requirement

Building Regulations 2010 – Schedule 1, Part C

The relevant functional requirement is that the building shall be constructed so as to:

> **"adequately resist the passage of moisture and groundwater."**

The Building Regulations are **outcome-based**, not prescriptive. They do not specify equipment counts or minimum components. Compliance is assessed by whether the adopted solution **functions effectively under foreseeable conditions**.

A system that is known to fail under foreseeable conditions **does not satisfy Part C**, irrespective of whether it represents a lower-cost or simplified installation.

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3. Approved Document C

Approved Document C provides statutory guidance on how Part C may be satisfied. In relation to below-ground and partially below-ground structures, it:

- Recognises **Type C (drained) waterproofing systems** as an acceptable method
- Requires systems to be appropriate for **site conditions**, groundwater risk, and long-term performance
- **Defers explicitly to British Standards** for design methodology

Approved Document C does **not** state that: - A single pump is sufficient - Redundancy is unnecessary - Foreseeable pump or power failure may be disregarded

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4. BS 8102:2022 – Governing Design Standard

BS 8102:2022 is the recognised British Standard referenced by Approved Document C for the protection of below-ground structures against water ingress.

BS 8102 requires: - A **risk-based design approach** - Consideration of the **likelihood and consequences of failure** - Avoidance of **single-point failure** where consequences are significant - Systems that remain effective and maintainable over their service life

For **Type C drained systems**, BS 8102 recognises that reliance on mechanical components introduces inherent failure risk and that redundancy is required where failure would result in internal flooding.

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5. Dual Pumps (Duty / Standby)

A single-pump installation represents a single point of failure. Failure of the pump, float switch, or blockage results in total system failure.

In a partially below-ground residential property, such failure would foreseeably result in internal flooding. Applying BS 8102 principles, this outcome is unacceptable.

> **Dual pumps are therefore required to avoid single-point failure and to achieve compliance with the functional requirements of Part C.**

This configuration reflects established industry practice for compliant Type C waterproofing systems.

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6. Battery Backup

Groundwater ingress and flooding risk are foreseeably associated with storm conditions, which are also commonly associated with **power outages**.

A pumped system that relies solely on mains power and ceases to function during these conditions **cannot be said to adequately resist groundwater**, as required by Part C.

BS 8102 requires designers to consider system performance during power failure.

> **Battery backup is therefore necessary to ensure continued system operation under foreseeable conditions and to achieve regulatory compliance.**

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7. Separate Internal and External Sump Chambers

The separation of internal Type C drainage and external groundwater management: - Reduces hydraulic overload risk - Prevents cascading system failure - Aligns with BS 8102 principles of robust, layered protection

Combining these systems would increase the consequences of failure and undermine compliance.

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8. Expert Evidence

- **Maclennan** designed and quoted the system incorporating dual pumps, battery backup, and system separation
- **Adam Clutterbuck**, an independent waterproofing specialist, reviewed and endorsed this approach
- He confirmed that redundancy and battery backup are **essential to effective system operation**, not optional enhancements

No equivalent expert evidence has been provided to support a single-pump, non-redundant alternative.

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9. Conclusion

There is **no support** within the Building Regulations, Approved Document C, or BS 8102 for reliance on a single pump without redundancy or backup power.

Applying the regulatory framework correctly leads to the following conclusion:

> **Dual pumps, battery backup, and separated sump systems are required to achieve Building Regulations compliance in a partially below-ground property. These measures constitute necessary reinstatement, not betterment.**

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